ALDEGID VA KETONLAR

Karbonil guruhli birikmalar · R–CHO va R–CO–R’

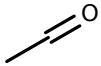
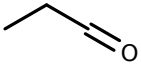
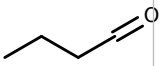
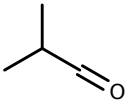
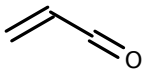
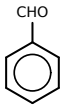
Student book – nazariy qo‘llanma

Molekulasida **karbonil guruhi >C=O** bo‘lgan organik birikmalar **karbonil birikmalar** deyiladi. **Aldegidlarda** karbonil uglerod bir tomondan **vodorod** bilan bog‘langan (**–CHO**, aldegid guruhi), **ketonlarda** esa ikkala tomondan **uglevodorod radikallari** bilan (R–CO–R’). To‘yingan aldegid va ketonlarning umumiy formulasi $\mathbf{C_nH_{2n}O}$; $M_r = 14n + 16$.

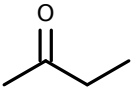
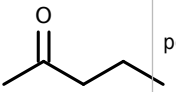
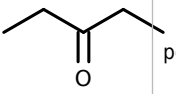
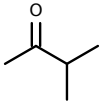
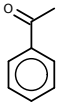
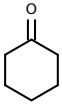
Muhim: bir xil $C_nH_{2n}O$ tarkibli aldegid va keton – o‘zaro **sinflararo izomerlar** ($n \geq 3$). Skelet formulalarda aldegid guruhidagi vodorod ko‘rsatilmagan.

1. Tasnifi va nomlanishi

A) Aldegidlar

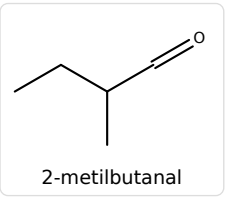
Formula	Skelet	Sistematik nomi	Tarixiy nomi
HCHO		metanal	formaldegid, chumoli aldegidi
CH_3CHO		etanal	asetaldegid, sirka aldegidi
C_2H_5CHO		propanal	propion aldegidi
C_3H_7CHO		butanal	moy aldegidi
$(CH_3)_2CH–CHO$		2-metilpropanal	izomoy aldegidi
$CH_2=CH–CHO$		propenal	akrolein
$C_6H_5–CHO$		benzaldegid	achchiq bodom moyining aldegidi

B) Ketonlar

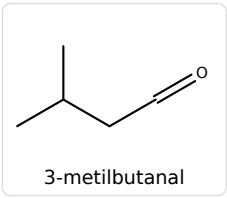
Formula	Skelet	Sistematik nomi	Tarixiy nomi
$CH_3–CO–CH_3$		propanon	aseton, dimetilketon
$CH_3–CO–C_2H_5$		butanon	metiletilketon
$CH_3–CO–C_3H_7$		pentanon-2	metilpropilketon
$C_2H_5–CO–C_2H_5$		pentanon-3	dietilketon
$CH_3–CO–CH(CH_3)_2$		3-metilbutanon-2	metilizopropilketon
$C_6H_5–CO–CH_3$		1-feniletanon	asetofenon
$C_6H_{10}=O$		siklogeksanon	–

C) Nomlash qoidalari

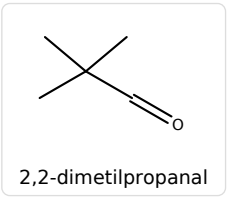
- **Aldegid:** alkan nomiga **«-al»** qo‘shiladi (etan → etanal). Zanjir **–CHO uglerodidan** raqamlanadi; u doim **1-raqam**, uni ko‘rsatish shart emas.
- **Keton:** alkan nomiga **«-on»** qo‘shiladi (propan → propanon). Raqamlash karbonil guruhga **eng yaqin uchidan** boshlanadi, karbonil o‘rni oxirida raqam bilan yoziladi (pentanon-2).
- Eng uzun zanjir **karbonil uglerodni** o‘z ichiga olishi kerak; radikallar alifbo tartibida, raqamlari bilan yoziladi.
- Aldegidning **tarixiy nomi** mos karbon kislota nomidan olinadi: chumoli kislota → **formaldegid**, sirka kislota → **asetaldegid**.



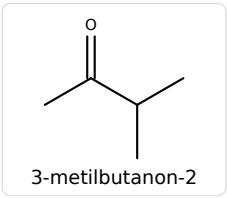
2-metilbutanal



3-metilbutanal



2,2-dimetilpropanal



3-metilbutanon-2

2. Tuzilishi

- Karbonil uglerod **sp^2-gibridlangan**: guruh atrofida **yassi uchburchak** tuzilish, valent burchak ≈ 120°.
- C=O qo‘shbog‘ida **1 ta σ- va 1 ta π-bog‘** bor. Kislorod uglerodga qaraganda **elektrmanfiyroq**, shuning uchun bog‘ **kuchli qutblangan**: uglerod atomi qisman musbat (δ+), kislorod qisman manfiy (δ–).
- Uglerodning musbat zaryadi tufayli karbonil uglerodga **nukleofil** reagentlar hujum qiladi – shuning uchun aldegid va ketonlar uchun asosiy reaksiya turi **birikish**.
- Aldegid guruhidagi **C–H bog‘** oson oksidlanadi: aldegidlar kuchli **qaytaruvchilar**. Ketonlarda bunday vodorod yo‘q – ular oksidlanishga chidamli.

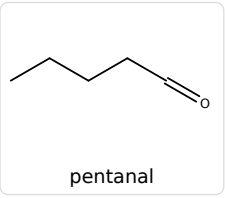
Modda (M_r ≈ 58–60)	Formula	t_{qayn}, °C
butan	C_4H_{10}	–1
aseton	CH_3COCH_3	56
propanol-1	C_3H_7OH	97
propion kislota (M = 74)	C_2H_5COOH	141

Aldegid va ketonlar molekulalari orasida **vodorod bog‘ hosil bo‘lmaydi** (O–H guruhi yo‘q): qaynash harorati mos spirtlardan **past**, alkanlardan esa qutbliligi tufayli **yuqori**.

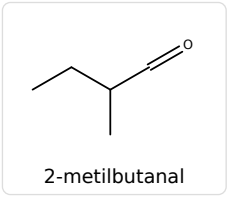
3. Izomeriyasi

Aldegid va ketonlarda **uglerod skeleti**, **karbonil guruhning o‘rni** (faqat ketonlarda) va **sinflararo** izomeriya uchraydi.

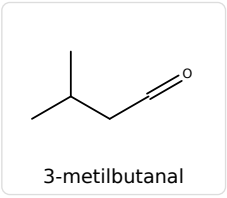
$C_5H_{10}O$ tarkibli karbonil birikmalar – **7 ta** izomer: 4 ta aldegid va 3 ta keton:



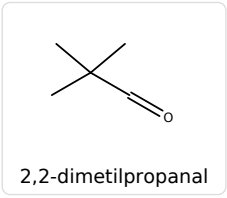
pentanal



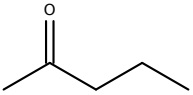
2-metilbutanal



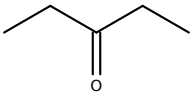
3-metilbutanal



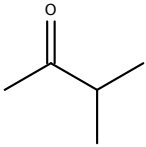
2,2-dimetilpropanal



pentanon-2



pentanon-3



3-metilbutanon-2

C_3H_6O: propanal (aldegid) va propanon (keton). **C_4H_8O** tarkibli karbonil birikmalar – 3 ta: butanal, 2-metilpropanal, butanon.

4. Fizik xossalari

- **Formaldegid** (t_{qayn} = –19 °C) – **o‘tkir hidli gaz**; **asetaldegid** (t_{qayn} = 21 °C) – past haroratda qaynaydigan suyuqlik.
- C_3–C_{16} aldegid va ketonlar – **suyuqliklar**, yuqorilari – qattiq moddalar. Aseton (t_{qayn} = 56 °C) – **o‘ziga xos hidli suyuqlik**.
- Quyi a’zolari (HCHO, CH_3CHO, aseton) suv bilan **istalgan nisbatda aralashadi** (suv bilan vodorod bog‘ hosil qiladi). Radikal ortishi bilan eruvchanlik kamayadi.
- **Formalin** – formaldegidning 40 % li suvli eritmasi (konservant, dezinfeksiya).
- Aseton – ko‘p organik moddalarni yaxshi eritadigan **erituvchi**.

Ehtiyot bo‘ling: formaldegid **zaharli** (shilliq pardalarni qitiqlaydi); aseton va asetaldegid **tez alangalanuvchi**.

5. Olinishi

1) Spirtlarni oksidlash (CuO yoki Cu, t)

Birlamchi spirtlardan – **aldegid**, **ikkilamchi** spirtlardan – **keton** hosil bo‘ladi (uchlamchi spirtlar oksidlanmaydi).

